PROJECT NAME

Smart Parking Allocation System Management



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WORKBOOK TEMPLETE

Project Name:

S-PARK

Smart Parking Allocation System

Problem Statement:

Finding an available parking space is often time-consuming and inconvenient. The proposed **SPARK (Smart Parking Allocation & Reservation System)** helps users find and reserve available parking slots online, reducing parking search time and avoiding slot conflicts.

Why does this problem exist?

No proper system to show available and occupied parking slots.
Drivers waste time searching for empty spaces.
Parking information is often not updated in real time.
Manual parking allocation can cause confusion and double booking.
Lack of an online booking system makes parking less convenient.
Limited parking space is not managed efficiently.

Who are the stakeholders?

Internal Stakeholders:

- Parking Administrator – manages parking slots and bookings.
- Parking Staff – monitors vehicles and parking areas.
- Organization/Institute Management – manages overall parking facilities.
- System Administrator/Developer – maintains the website and system.

External Stakeholders:

- Vehicle Owners/Users – search and book parking slots.
- Visitors/Guests – use parking facilities temporarily.
- Parking Service Provider – manages parking services, if applicable.

Expected system users:

- Registered Users/Vehicle Owners – search and book available parking slots.
- Visitors – check parking availability and make bookings.
- Parking Staff – monitor slot occupancy and manage vehicle entry/exit.
- Administrator – manage users, parking slots, bookings, and system records.
- Management – view parking usage and overall system reports.

Objective of the project:

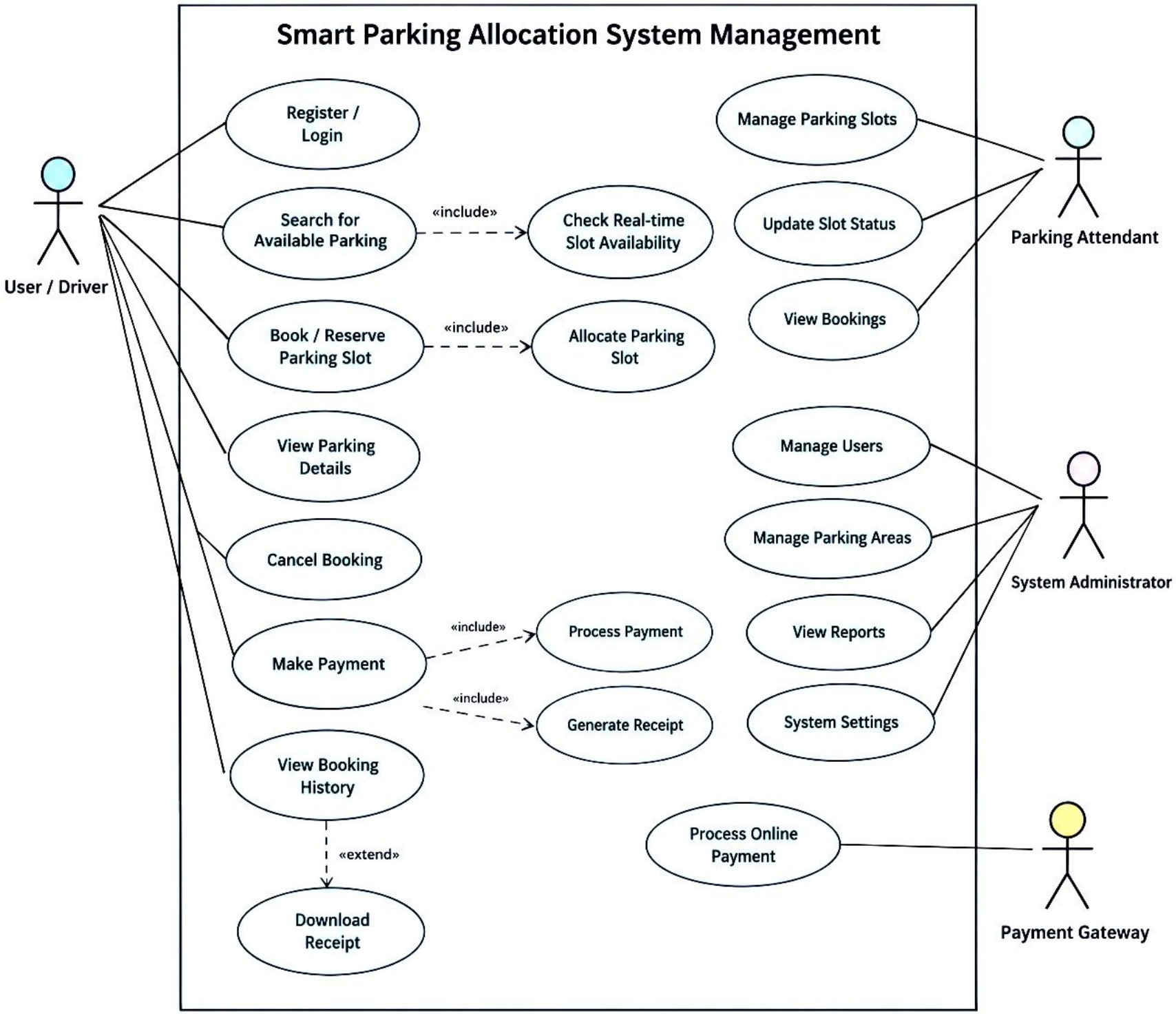
- To develop an **online smart parking allocation system**
- To help users **find available parking slots easily**
- To allow users to **reserve parking slots online**
- To reduce the **time spent searching for parking**
- To avoid **double booking and parking conflicts**
- To help administrators **manage parking slots efficiently**
- To maintain **accurate parking and booking records**

Expected outputs:

- Display **available, occupied, and reserved parking slots**
- Provide **online parking slot booking and allocation**
- Generate **booking confirmation with Slot ID/Booking ID**
- Show **user booking and parking history**
- Provide an **admin dashboard** for managing slots and bookings.
- Maintain **accurate and updated parking records**
- Reduce **parking search time and slot conflicts**

Hands-on-Activity:

- Conducted a **survey** to identify common parking problems faced by users.
- Collected information about **parking areas, available slots, and vehicle types**
- Prepared a **data-gathering table** with required fields and data types.
- Designed a basic **website interface/wireframe** for parking allocation.
- Created a **parking slot layout** showing Available, Occupied, and Reserved slots.
- Prepared the basic **system workflow** from parking search to slot booking.
- Identified the **stakeholders and expected system users**
- Discussed how **IoT sensors** could be used for automatic slot detection in future.


Smart Parking Allocation System Management
User / Driver
Register / Login
Search for Available Parking
«include»
Check Real-time Slot Availability
Book / Reserve Parking Slot
«include»
Allocate Parking Slot
View Parking Details
Cancel Booking
Make Payment
«include»
Process Payment
«include»
Generate Receipt
View Booking History
«extend»
Download Receipt
Manage Parking Slots
Update Slot Status
View Bookings
Parking Attendant
Manage Users
Manage Parking Areas
View Reports
System Settings
System Administrator
Process Online Payment
Payment Gateway

Smart parking allocation system management

1. Descriptive Analysis

- Describes the current parking situation and available parking spaces.
- Shows the number of available, occupied, and reserved slots.
- Provides information about parking usage, bookings, and vehicle details.

2. Diagnostic Analysis

- Identifies why parking problems occur, such as limited spaces or improper allocation.
- Analyzes frequent slot occupancy and booking conflicts.
- Helps administrators understand the causes of parking-related issues.

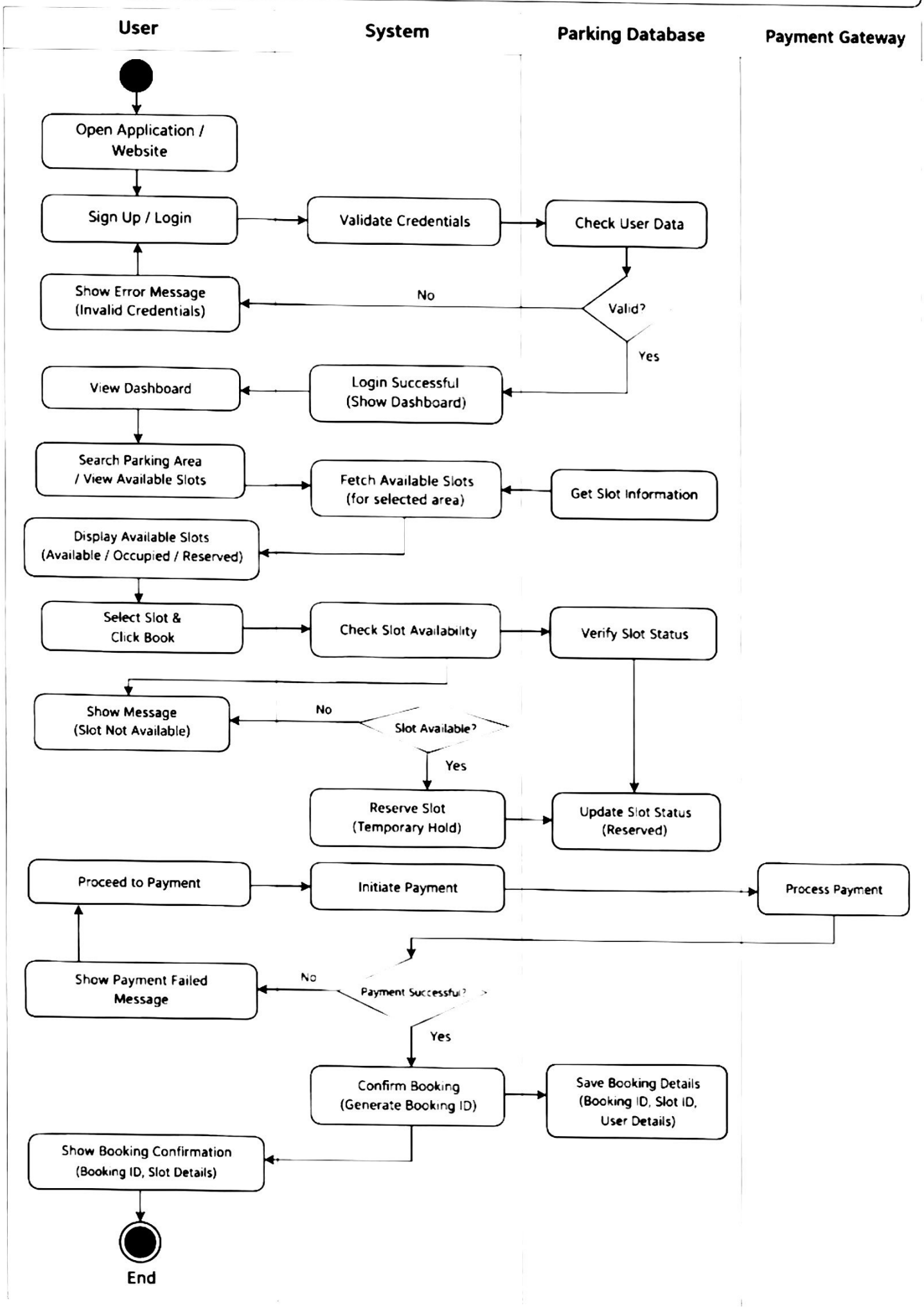
3. Predictive Analysis

- Uses previous parking and booking data to predict future parking demand.
- Can identify busy periods when parking spaces are likely to be highly occupied.
- Helps users and administrators plan parking availability in advance.

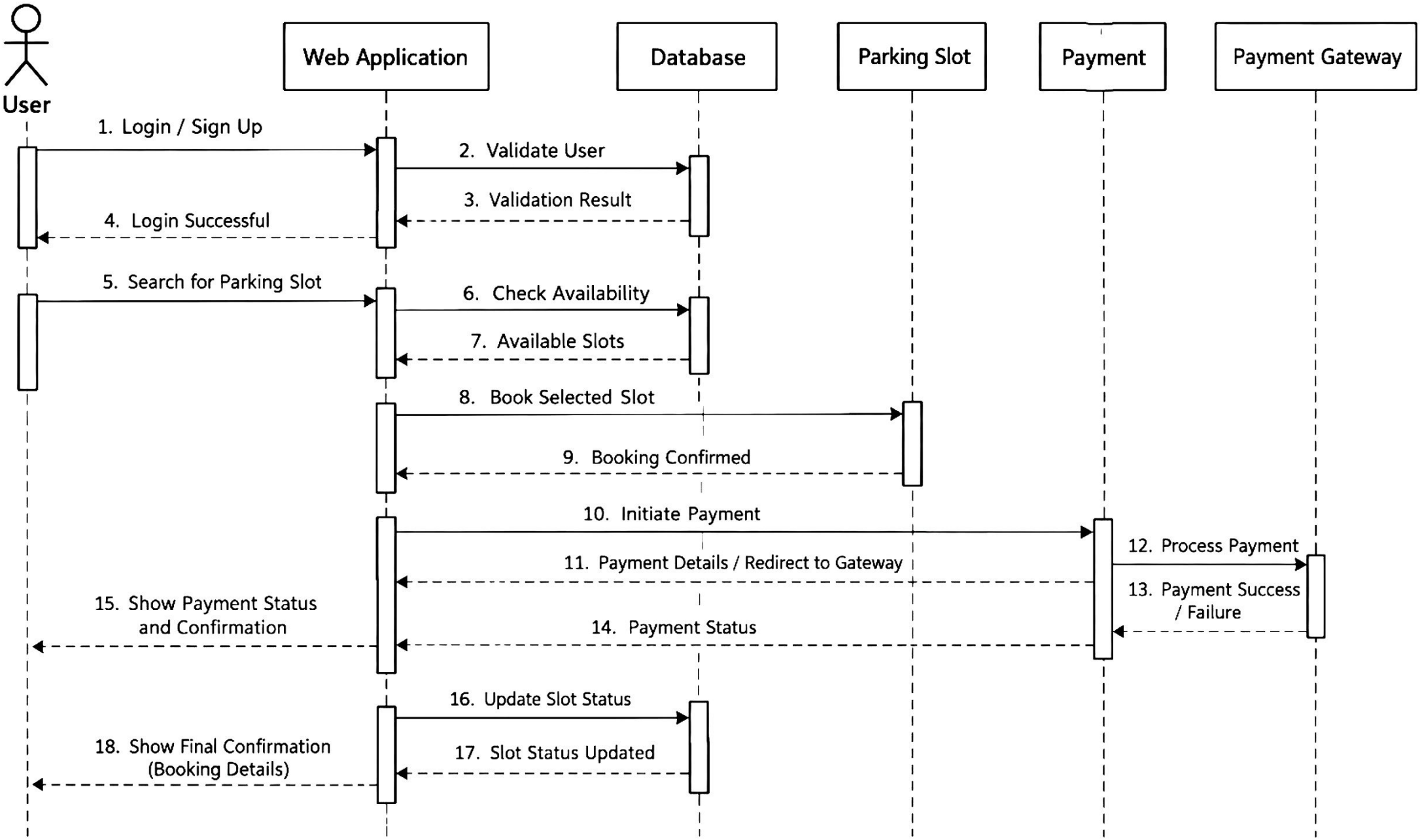
4. Prescriptive Analysis

- Suggests the best available parking slot based on availability and user requirements.
- Helps administrators allocate parking spaces efficiently.
- Can recommend actions such as reserving suitable slots during high-demand periods.

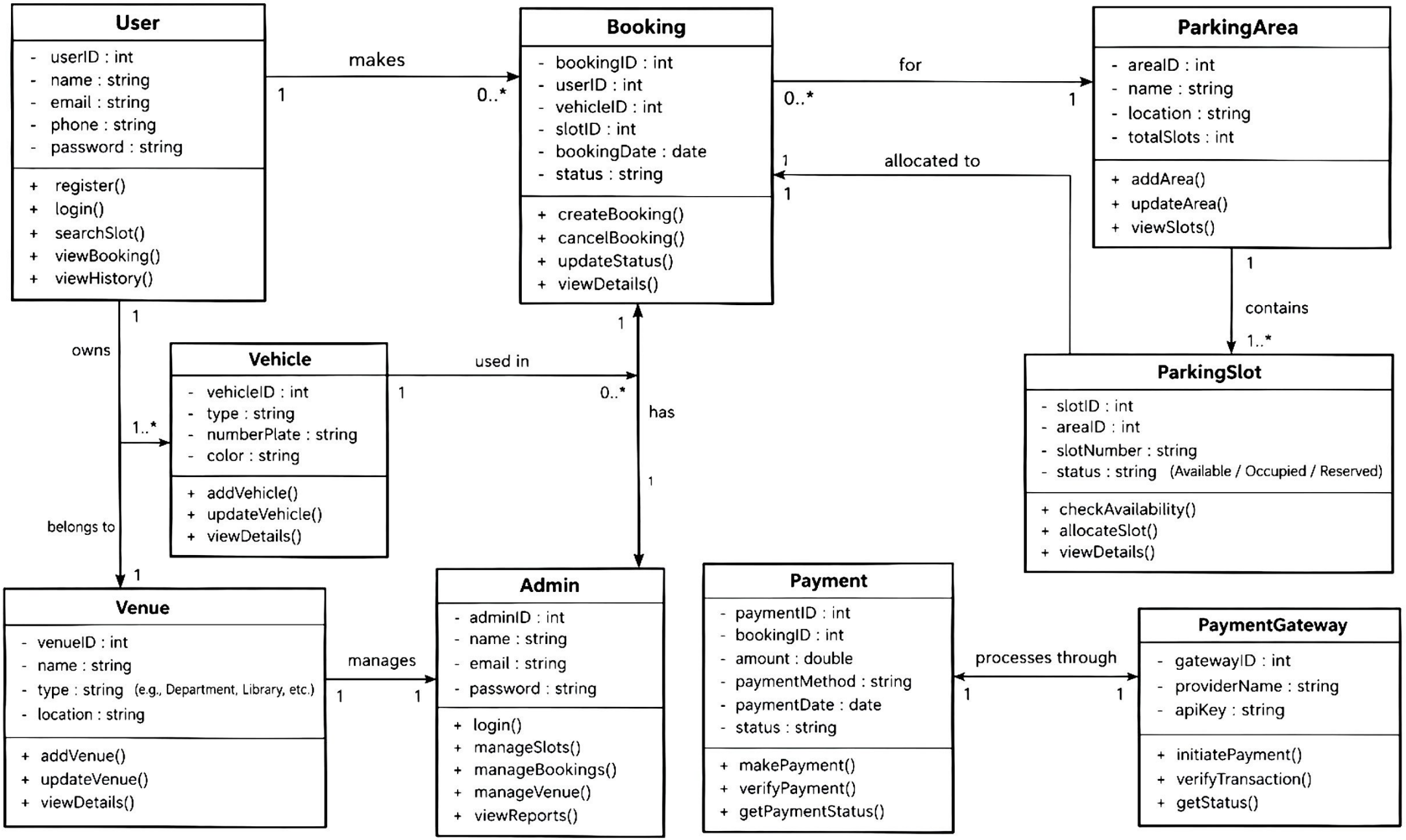
Activity Diagram – SPARK (Smart Parking Allocation & Reservation System)



Sequence Diagram – Online Parking Allocation System
(with Payment and Payment Gateway)



Online Parking Allocation System – Class Diagram



Collaboration Diagram – Online Parking Allocation System

